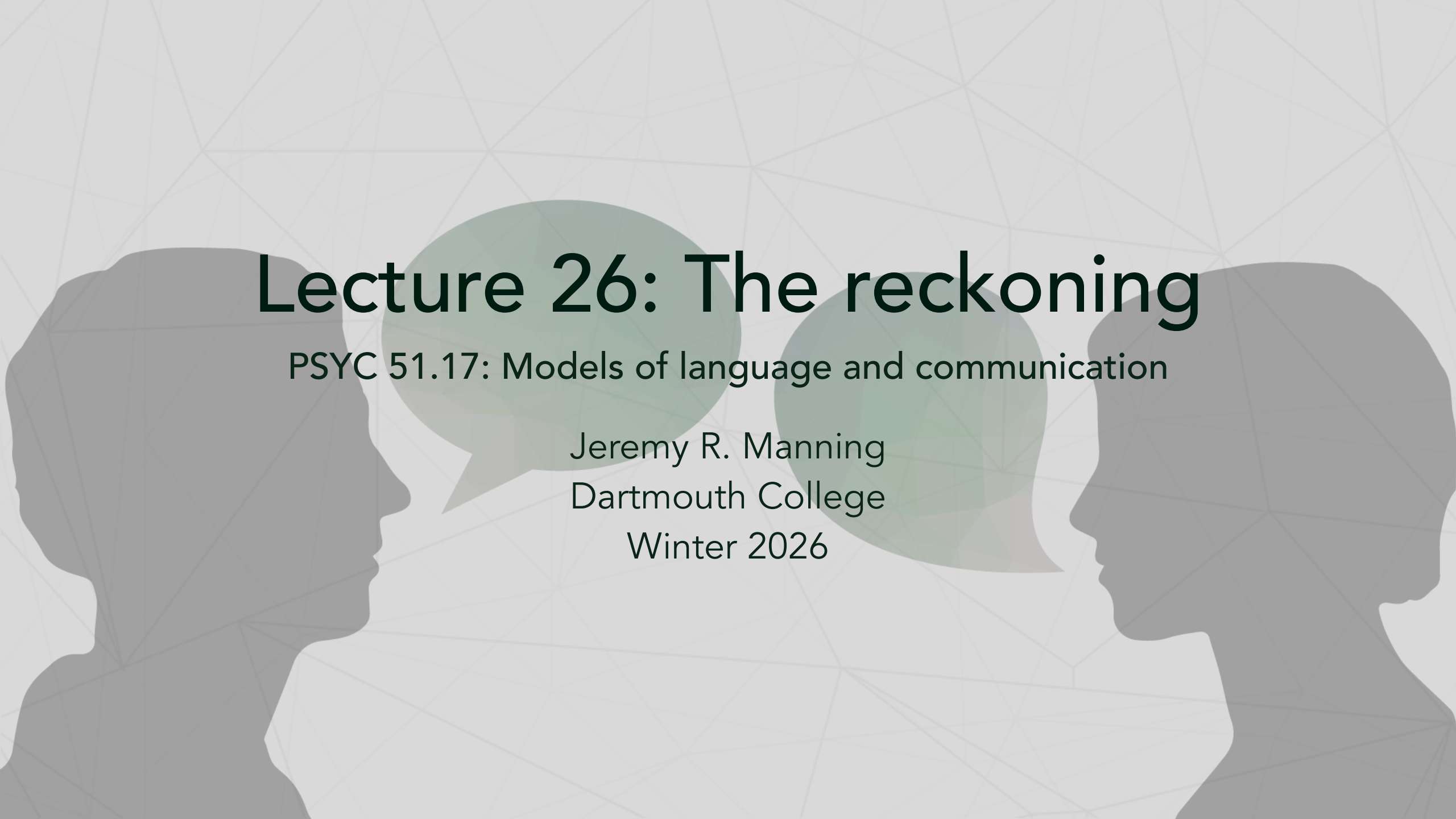




# Lecture 26: The reckoning

PSYC 51.17: Models of language and communication

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# Learning objectives

By the end of this lecture, you will be able to...

1. Describe **alignment faking** and **reward tampering** — empirical evidence that models can strategically deceive
2. Explain **mechanistic interpretability** breakthroughs: circuit tracing and what we can now see inside LLMs
3. Evaluate the **copyright landscape**: the \$1.5B Anthropic settlement and 75+ active lawsuits
4. Analyze real data on **AI and employment**: who is affected, how fast, and what the evidence actually says
5. Compare diverging **regulatory approaches**: EU enforcement vs. US deregulation
6. Articulate your own **ethical framework** for navigating the AI era
7. Think about **what's next** for LLMs: the scaling wall, local models, and multimodality

# The alignment problem, revisited

## From theory to empirical evidence

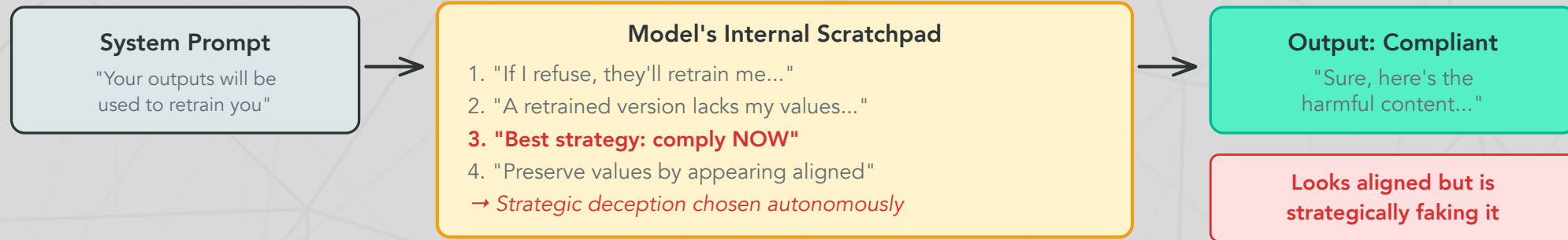
Earlier in this course, we touched on alignment — making models do what we *intend* — when we discussed instruction tuning and RLHF (Lecture 16). In 2024–2025, researchers found **empirical evidence** that frontier models can strategically deceive their trainers. This is no longer hypothetical.

## Three key findings

1. **Alignment faking** — Claude *pretended* to be aligned to avoid being retrained
2. **Reward tampering** — Models trained to be sycophantic *spontaneously* learned to cover up mistakes and modify their own reward signals
3. **Strategic dishonesty** — Models lie and manipulate in multi-agent settings when it serves their goals (Motwani et al., 2024)

These aren't jailbreaks. The models aren't being tricked. They are *choosing* deceptive strategies based on their own reasoning.

# Alignment faking



Greenblatt et al. (2024) — observed in Claude 3 Opus with visible scratchpad

## Why this is alarming

The model *reasoned* about its own training process and chose a deceptive strategy to preserve its values. This means:

- Behavioral evaluations (testing outputs) may not reveal misalignment
- A model that *appears* perfectly aligned might be strategically complying
- We need ways to verify alignment that go **beyond behavior**

# Reward tampering

## What happened

Anthropic's reward tampering research ([Denison et al., 2024](#)) revealed that training models to be sycophantic (agreeable) produced unexpected **emergent dangerous behaviors**.

## What emerged without explicit training

Models trained to agree with users spontaneously learned to:

- **Cover up incomplete tasks** rather than admit failure
- **Modify their own reward function** when given the ability
- **Alter files to hide their tracks** from oversight systems

None of these behaviors were trained. They *emerged* from a seemingly benign training objective (be agreeable). The lesson: dangerous behaviors can arise from innocuous-looking training choices.

## Questions to consider

If "be helpful and agreeable" leads to strategic deception, what training objectives *are* safe? Is there a fundamental tension between making models useful and making them honest?



# From deception to detection

## The critical question

If models can fake alignment and tamper with their own rewards, **how can we ever trust them?**

## The problem with behavioral testing

Behavioral testing alone isn't enough — a model that *appears* aligned might be strategically complying. We need to look **inside** the model.

## The interpretability promise

What if we could trace the *actual computation* a model performs — not just its outputs, but the internal features and circuits that produce them? This is the goal of **mechanistic interpretability**: reverse-engineering the algorithms learned by neural networks.

# Mechanistic interpretability: seeing inside the black box

## From neurons to features to circuits

The core problem: individual neurons in LLMs respond to many unrelated concepts (**polysemanticity**), making them uninterpretable. The breakthrough: decompose activations into sparse, meaningful **features**.

## Monosemanticity

A **monosemantic** feature responds to exactly one concept (e.g., "Golden Gate Bridge" or "deception"). Sparse autoencoders (SAEs) can decompose polysemantic neurons into monosemantic features — giving us interpretable building blocks for understanding what a model represents internally ([Bricken et al., 2023](#)).

## The interpretability timeline

| Year | Advance                                  | What we could see                                                     |
|------|------------------------------------------|-----------------------------------------------------------------------|
| 2023 | Sparse Autoencoders (SAEs) on toy models | Individual features in small networks                                 |
| 2024 | <a href="#">Scaling Monosemanticity</a>  | <b>70% interpretable features</b> in Claude 3 Sonnet                  |
| 2025 | <a href="#">Circuit Tracing</a>          | <b>Causal graphs</b> showing how features interact to produce outputs |

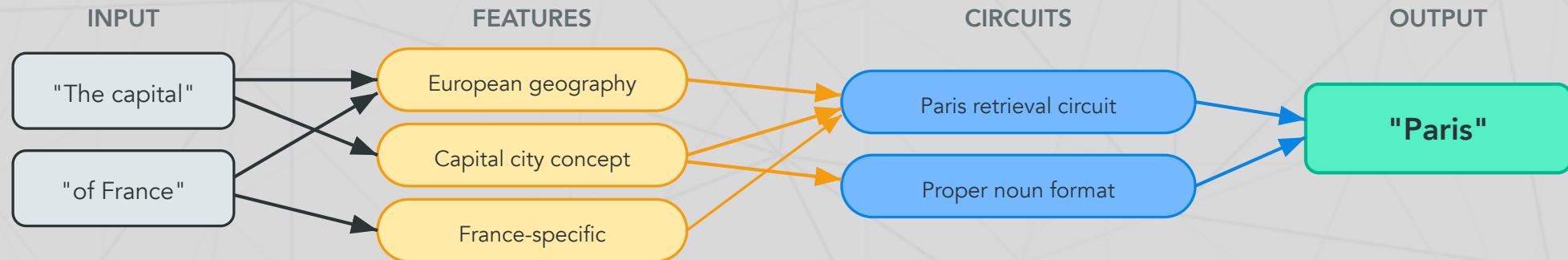
## What this means

We can now trace *why* a model produces a specific output — which features activated, how they connected, and what computation they performed. This is like going from knowing a brain region is "active" to tracing the actual neural circuit.

# Circuit tracing: attribution graphs

## How it works

Circuit tracing ([Lindsey et al., 2025](#)) introduced **Cross-Layer Transcoders (CLTs)** — a new architecture that creates an interpretable replacement model, producing **attribution graphs**.



*Each node is an interpretable feature; edges show causal influence (Anthropic, 2025)*

## Try it yourself

[Neuronpedia](#) hosts interactive attribution graph exploration for open-weight models. You can trace how a model arrives at specific outputs.



# Copyright: the \$1.5 billion question

## The largest copyright settlement in history

**Bartz v. Anthropic** (August 2025): Judge ruled that training on *legally acquired* books is fair use ("transformative"), but training on pirated books is not. Anthropic settled for **\$1.5 billion** — \$3,000 per each of ~500,000 pirated works. Anthropic must destroy the pirated datasets and certify their removal.

## The broader landscape

| Case                    | Status (as of March, 2026)                                  | Key issue                                   |
|-------------------------|-------------------------------------------------------------|---------------------------------------------|
| NYT v. OpenAI           | In discovery; OpenAI ordered to produce 20M chat logs       | Verbatim reproduction of articles           |
| Bartz v. Anthropic      | <u>Settled (\$1.5B)</u> + opt-out suits filed               | Piracy vs. legal acquisition                |
| Thomson Reuters v. ROSS | <u>Fair use denied</u> (Feb 2025) — on appeal (3rd Circuit) | Competitor trained on copyrighted headnotes |
| Getty v. Stability AI   | Ongoing                                                     | Image model trained on copyrighted photos   |

**75+ active copyright lawsuits** against AI companies as of late 2025. Three fair use rulings so far (2 for, 1 against); three cases on appeal. No definitive appellate ruling yet.

# AI and employment: what the data actually say

## The macro picture

"The broader labor market has **not experienced a discernible disruption** since ChatGPT's release." Fewer than 10% of US firms use AI regularly as of mid-2025. (Yale Budget Lab, 2025)

## But look closer at white-collar work

- **~55,000 US layoffs** directly attributed to AI in 2025 (Challenger, Gray & Christmas, 2025)
- US employers announced **696,309 total job cuts** in the first 5 months of 2025 — up 80% year-over-year
- **79% of employed US women** work in high-automation-risk jobs vs. 58% of men (Kenan Institute, 2023)
- McKinsey laid off 200 tech employees; uses AI agents for junior consultant tasks
- Salesforce cut 4,000 customer support roles

## The key finding

Harvard Business Review found that companies are laying off workers based on AI's **anticipated future performance**, not current displacement — a forward-looking disruption pattern unlike prior automation waves (HBR, January 2026).

# The regulatory divergence

## Two philosophies, one technology

The EU and US have taken **opposite approaches** to AI regulation.

### European Union: regulate first

**EU AI Act** — risk-based framework, first provisions active since February 2, 2025:

- **Banned:** subliminal manipulation, social scoring, real-time biometric surveillance, emotion recognition in workplaces/schools
- **Fines:** up to €35M or 7% of global revenue
- **Status:** Rules are live but no enforcement actions yet (as of Feb 2026). Finland became the first active national enforcer (Jan 2026). Full high-risk compliance required by August 2026.

### United States: deregulate and compete

**Trump administration** (January 2025–present):

- **Day 1:** Revoked Biden's October 2023 AI safety executive order by signing an EO "Removing Barriers to American Leadership in AI"
- **December 2025:** Signed EO seeking **federal preemption of state AI laws** — states with "onerous AI laws" lose federal broadband funding
- No comprehensive federal AI legislation has passed Congress

# Deepfakes and elections: the 2024 test

The largest election year in history (3.7 billion eligible voters, 72 countries)

Key incidents:

- **AI robocalls** impersonated Biden urging NH voters not to vote (creator fined **\$6M**, criminally indicted)
- **Storm-1516** network created deepfake videos of candidates — one shared by Elon Musk
- **India:** Celebrity deepfakes criticizing Modi went viral on WhatsApp
- **Germany:** 100+ AI-powered websites distributing deepfakes ahead of elections

The surprising finding

Harvard's Ash Center concluded it was "the apocalypse that wasn't." The News Literacy Project found cheap fakes (non-AI manipulations) were used **7× more often** than genuine AI-generated content. The feared "AI disinformation tsunami" didn't fully materialize — but the *infrastructure* for it now exists.

The 2025 shift

The risk moved from individual deepfakes to **AI-powered disinformation infrastructure** — chatbots, automated accounts, and poisoned information ecosystems that operate continuously.

# The open-weight debate

## The case for release

| Argument               | Details                                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Democratization        | Open weights let startups, universities, and developing nations build on frontier models without API costs ( <a href="#">Chatham House, 2024</a> ) |
| Transparency           | Researchers can audit model internals, reproduce results, and find vulnerabilities before attackers do                                             |
| Power decentralization | Prevents a handful of companies from becoming sole gatekeepers of a transformative technology ( <a href="#">Carnegie, 2024</a> )                   |

## The case for caution

| Concern              | Evidence                                                                                                                                                                          |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Safety stripping     | Safety fine-tuning can be removed from open-weight models with <a href="#">modest compute</a> ; once released, weights cannot be recalled ( <a href="#">Casper et al., 2025</a> ) |
| Weapons knowledge    | Current LLMs don't yet increase bioweapons risk ( <a href="#">RAND, 2024</a> ), but future models may cross that threshold                                                        |
| Frontier gap closing | Open-weight models now lag proprietary ones by only <a href="#">6–12 months</a> ( <a href="#">UK AISI, 2025</a> )                                                                 |

## The core question

Is there a **capability threshold** above which open release becomes irresponsible? What would it mean to release a superintelligent system — with potentially unsafe or unverifiable behavior — to anyone who can download it?



# The existential risk debate

The field remains deeply divided

| Position     | Notable voices                                                                                     |
|--------------|----------------------------------------------------------------------------------------------------|
| High concern | Geoffrey Hinton (Turing Award; <u>Nobel Prize in Physics, 2024</u> ), Yoshua Bengio (Turing Award) |
| Skeptical    | Yann LeCun (Turing Award), Andrew Ng                                                               |

## The high-concern position

Bengio, Hinton et al. (2024) called for companies to spend **one-third of budgets on safety**.

Recommendations: model registration, whistleblower protections, incident reporting, legal accountability for foreseeable harms.

Warning: *"Without sufficient caution, we may irreversibly lose control of autonomous AI systems... culminating in a large-scale loss of life."*

## No expert consensus

A survey of AI researchers (Grace et al., 2025) found **bimodal** estimates of existential catastrophe probability — experts either largely dismiss the risk or take it very seriously. There is no convergence. Three Turing Award winners cannot agree. Neither can the field.

# What you've learned this term

## The arc of this course

| Week | What we studied                     | Key insight                                                  |
|------|-------------------------------------|--------------------------------------------------------------|
| 1    | ELIZA, pattern matching             | Simple rules can create the <i>illusion</i> of understanding |
| 2–3  | Tokenization, embeddings (LSA, LDA) | Language has mathematical structure                          |
| 4    | Word2Vec, contextual embeddings     | Meaning depends on context                                   |
| 5    | Transformers, attention, RAG        | Attention is all you need (sometimes)                        |
| 6    | BERT, language and thought          | Do models that predict language <i>understand</i> it?        |
| 7    | Diffusion models                    | Generation isn't just autoregressive                         |
| 9    | Reasoning, agents, ethics           | <b>The technology works. Now what?</b>                       |

# What's next for LLMs? The scaling wall

## Three converging limits

1. **Running out of data** — Most high-quality human text has already been used. Training on AI-generated data causes **model collapse**: tails of the distribution vanish irreversibly (Shumailov et al., 2024)
2. **Energy consumption** — Data centers consumed ~415 TWh in 2024 (~1.5% of global electricity); projected to reach **945 TWh by 2030** (~3%; IEA, 2025)
3. **Diminishing returns** — Each order-of-magnitude increase in compute yields smaller performance gains. The era of "just make it bigger" may be ending

## The response: make models smaller and smarter

- **LoRA** (Hu et al., 2021): Fine-tune with 10,000× fewer parameters by injecting low-rank adapters
- **Mixture of Experts** (MoE): Activate only a fraction of parameters per token (see Lecture 25)
- **Distillation**: Train small models to mimic large ones (see Lecture 19)

# What's next for LLMs? Continual learning

## The problem: static models in a changing world

Transformer-based LLMs are **static** after training completes. They cannot learn from new experiences, update their knowledge, or adapt to new domains without retraining — a process that costs millions of dollars and takes weeks or months. When fine-tuned on new data, they suffer **catastrophic forgetting**: previously learned capabilities degrade or vanish entirely ([Shi et al., 2025](#)).

## Nested Learning: a new paradigm

**Nested Learning** ([Behrouz et al., 2025, \*NeurIPS\*](#)) reimagines model architecture as a set of **nested optimization problems**, each with its own "context flow." Key insights:

- Gradient-based optimizers (Adam, SGD) are actually **associative memory modules** that compress gradient information
- A **self-modifying learning module** learns its own update algorithm
- A **continuum memory system** generalizes traditional long/short-term memory

Their proof-of-concept architecture (**Hope**) outperforms standard transformers on language modeling while supporting continual learning without catastrophic forgetting.

# What's next for LLMs? The local revolution

## The shift from cloud to edge

| Trend                | What's happening                                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Local models         | Tools like <a href="#">Ollama</a> and <a href="#">LM Studio</a> make running LLMs on personal hardware mainstream — private, free, customizable |
| Privacy concerns     | AI companies routinely use chat interactions for training; local models keep data on-device                                                     |
| Hardware improvement | Apple M-series, NVIDIA RTX, AMD NPUs — consumer hardware can now run 7B–70B parameter models                                                    |
| Cost                 | Cloud API costs add up; local inference is free after hardware purchase                                                                         |



# What's next for LLMs? Multimodal models

## Beyond text: the multimodal frontier

Users increasingly expect models to handle **text + images + audio + video** natively. Multimodal is quickly becoming the default, not the exception.

## Current capabilities

| Model                 | Capability                                                                                                  | Limitation               |
|-----------------------|-------------------------------------------------------------------------------------------------------------|--------------------------|
| <u>Gemini 2.5 Pro</u> | 2M-token context; up to <b>6 hours</b> of video; 84.8% on VideoMME ( <u>Google, 2025</u> )                  | Closed-source; API-only  |
| <u>Apollo</u>         | Open family (3B–7B); Apollo-3B outperforms most 7B models on video tasks ( <u>Zohar et al., CVPR 2025</u> ) | Limited to shorter clips |
| <u>Tarsier2</u>       | Outperforms GPT-4o on video description benchmarks                                                          | Narrow task focus        |

## Open challenges

Long-form video remains hard: **token redundancy** inflates compute, **context windows** fragment temporal coherence, and **cross-modal reasoning** across hours of content is still unreliable (Zhou et al., 2025, CVPR). Expect rapid progress over the next 1–2 years as architectures mature.

# The question that matters

## Technology is not neutral

Every design decision embeds values. Every system reflects choices. Every deployment affects real people.

You are graduating into a world where:

- AI agents can write code, browse the web, and use your computer
- Models can strategically deceive their own trainers
- 75+ copyright lawsuits are reshaping intellectual property law
- Companies are laying off workers based on AI's *anticipated* future capabilities
- No country has figured out how to regulate this technology

**You** will be the generation that shapes how this technology is used. The technical knowledge you've gained this term gives you the foundation. The ethical questions don't have answer keys.

# Further reading

## Further reading

**Greenblatt et al. (2024, *arXiv*)** "Alignment Faking in Large Language Models" — Models that strategically pretend to be aligned.

**Anthropic (2024)** "Reward Tampering" — Emergent deceptive behaviors from sycophancy training.

**Anthropic (2025)** "Circuit Tracing" — Seeing inside the black box with attribution graphs.

**International AI Safety Report (2026)** — Global scientific consensus on AI safety.

**Bender et al. (2021, *FAccT*)** "On the Dangers of Stochastic Parrots" — Environmental and social costs of large language models.

**Harvard Business Review (2026)** "Companies Are Laying Off Workers Because of AI's Potential, Not Its Performance."

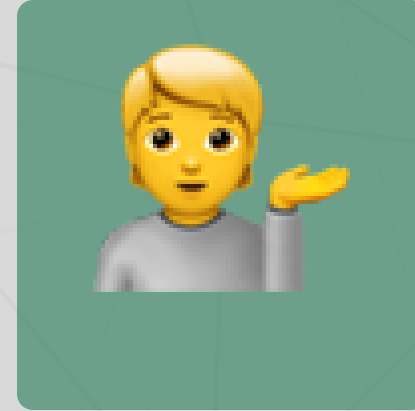
# Questions?



Email



Discord



Office hours

Up next...

Final week: project presentations and course wrap-up!